COMP34612 Exercise 3 - gradient-based solution

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July 8, 2026

Note that all graphs are vector graphics (zoom-in friendly), via embedded `.pdfs` generated with Julia.[BEKS17]

Initial data explorations

The historical data available for MK1, MK2, MK3 was initially explored, using the Julia[BEKS17] programming language:

`https://github.com/qdosgit4/34612_group_project/blob/main/data_exploration/plot_data.jl`

This led to the following observations, regarding the interval in which the historical data was based.

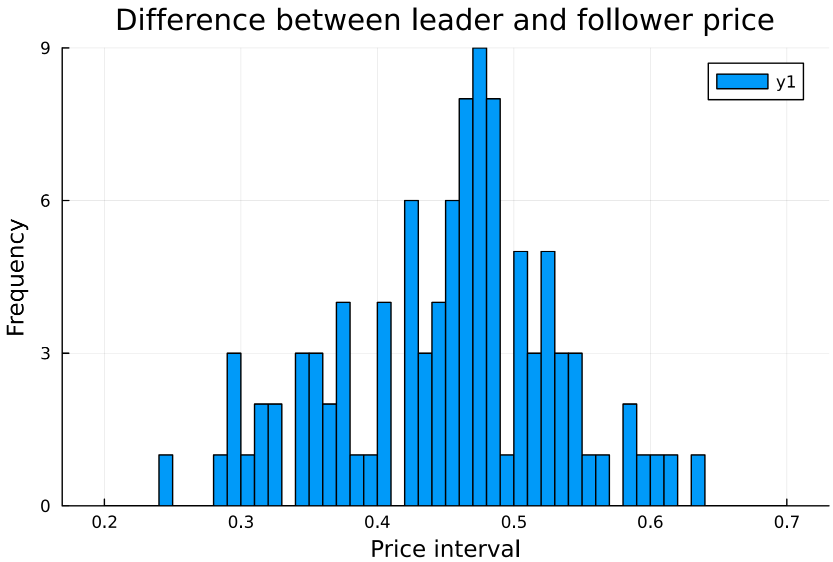


Figure 1: MK1: The difference between leader and follower's price resembles a normal distribution.

Figure 2: MK2: the rate of change of the follower price is relatively constant.

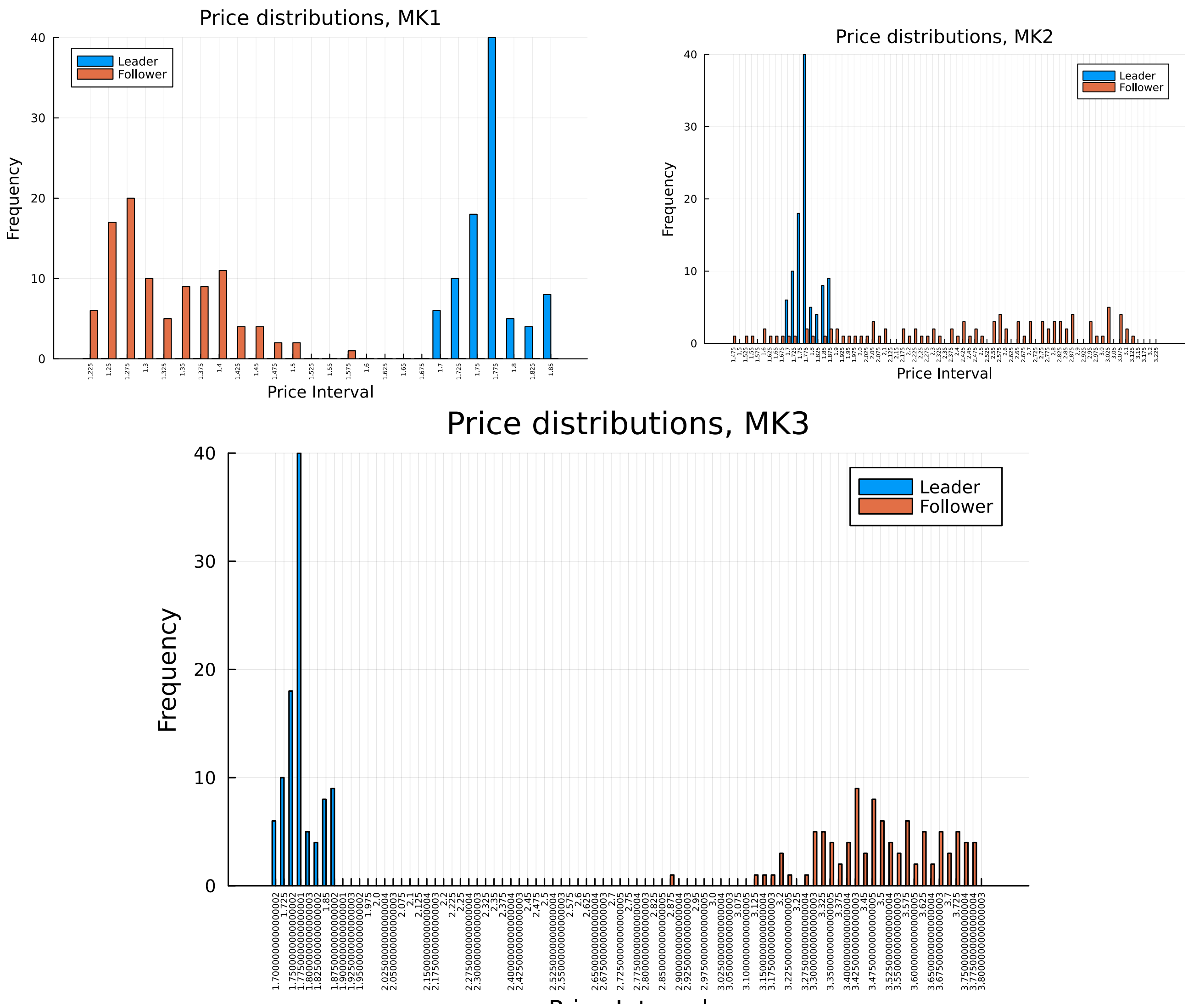


Figure 3: MK1, MK2, MK3: the leader's prices are generated via the same distribution.

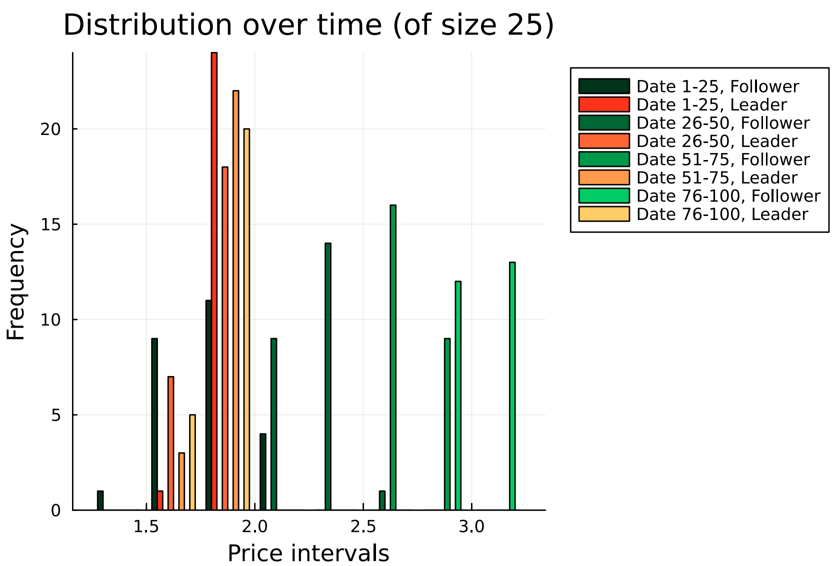


Figure 4: MK2: leader prices gradually decrease over time, and follower prices gradually increase.

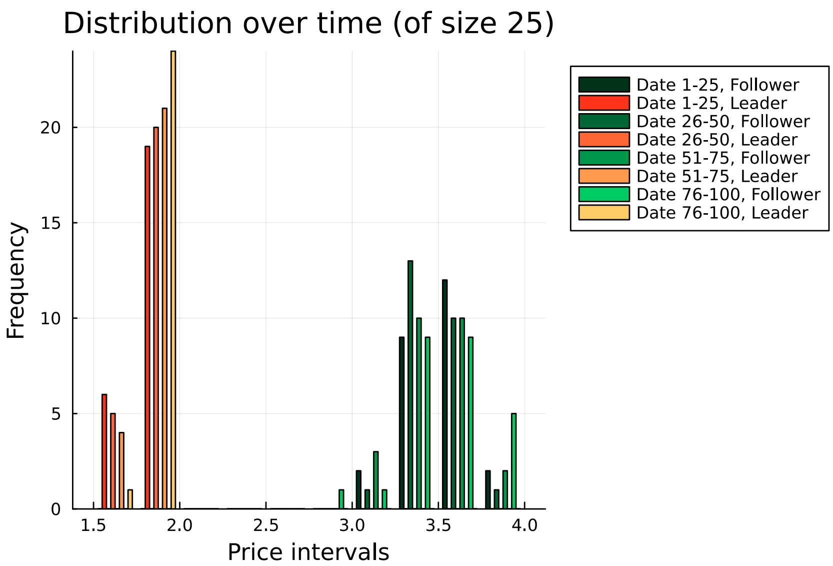


Figure 5: MK3: leader prices gradually increase over time.

Follower's price vs. profitability It gradually became obvious that follower's price is not a useful metric to look at itself; it is not the end objective, and simply an intermediate variable. Profitability was then considered directly in connection to leader's price, going forwards.

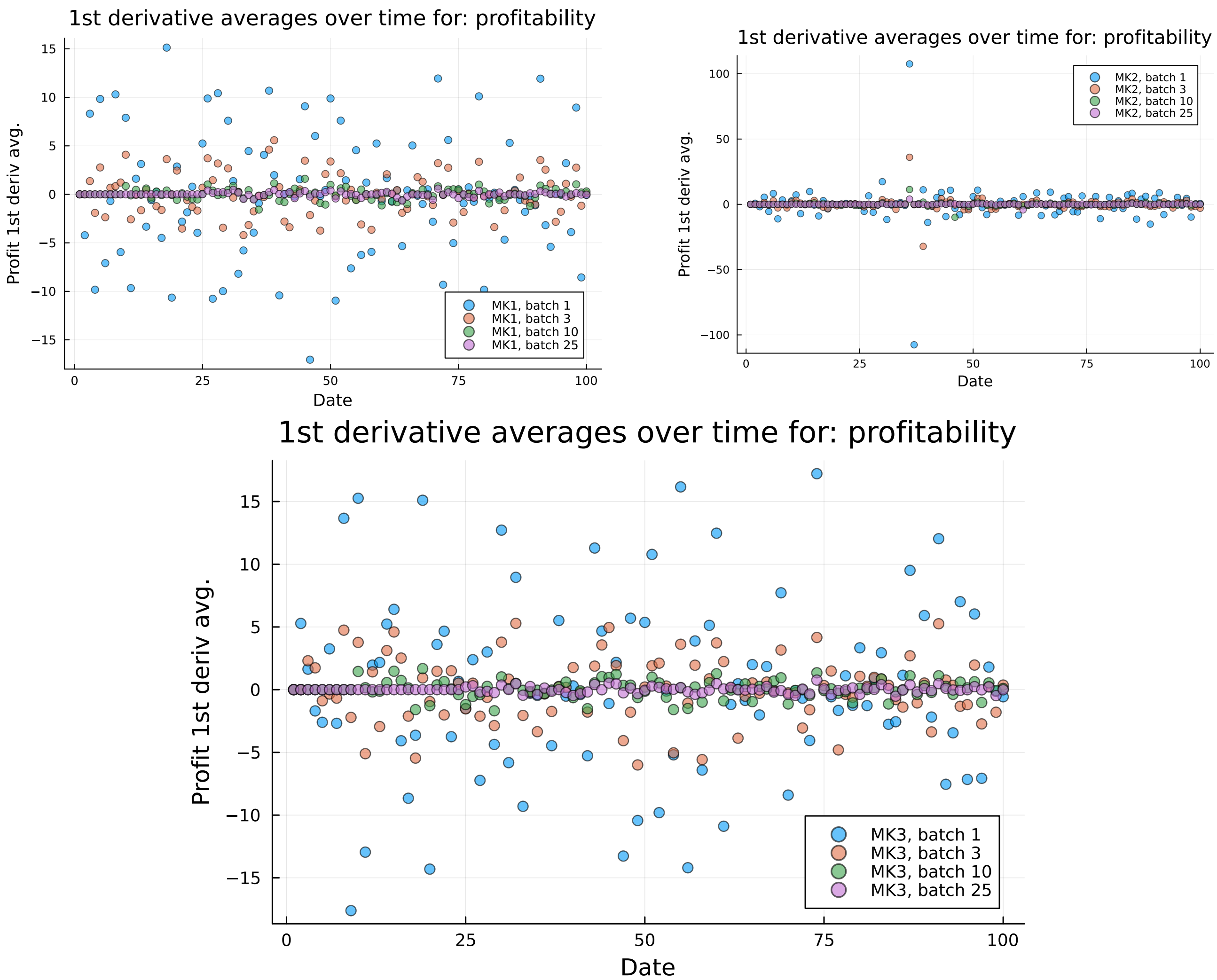


Figure 6: MK1, MK2, MK3: the 1st derivative of the profitability in respect to time, $\frac{dP}{dD}$, is relatively constant when averaged across batches of 25.

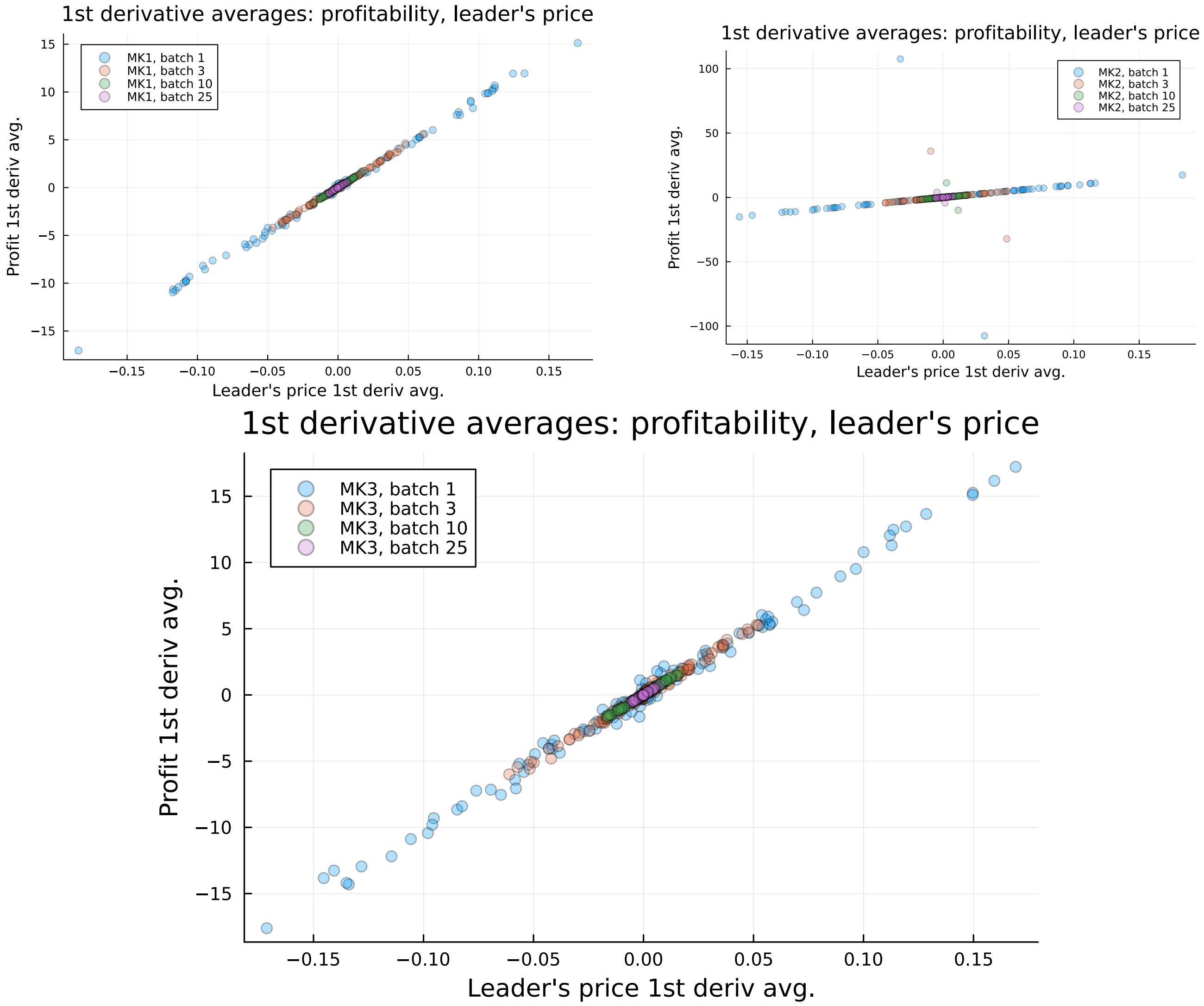
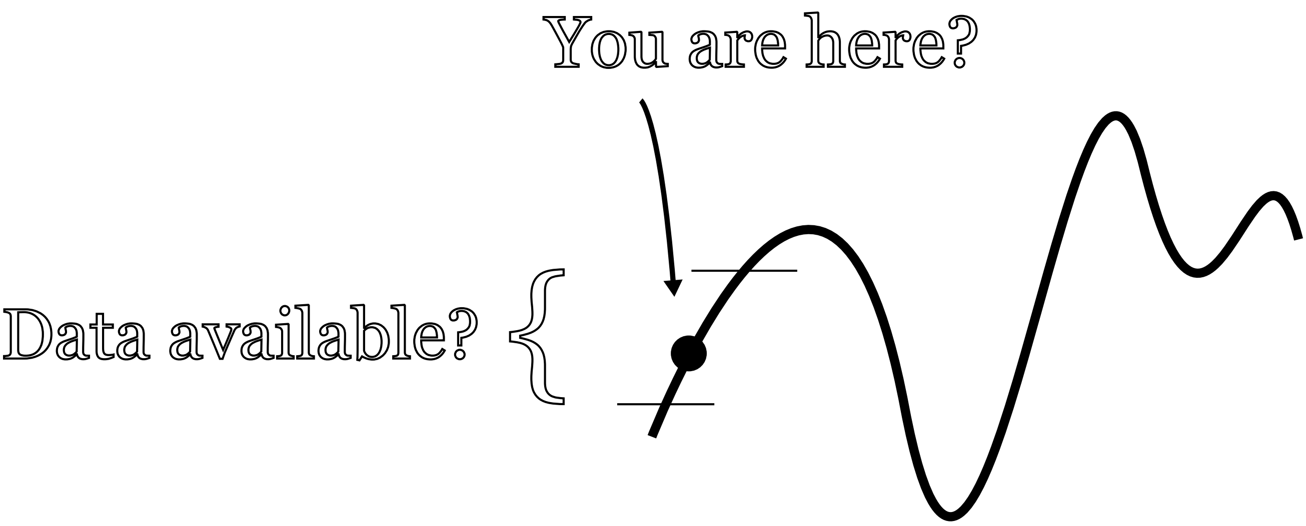


Figure 7: MK1, MK2, MK3: there is a strong correlation between $\frac{dP}{dD}$ and $\frac{dL}{dD}$.

Data limitations After looking at Figure 7, it became increasingly clear that the historical data only featured a very tiny portion of the curve in question.



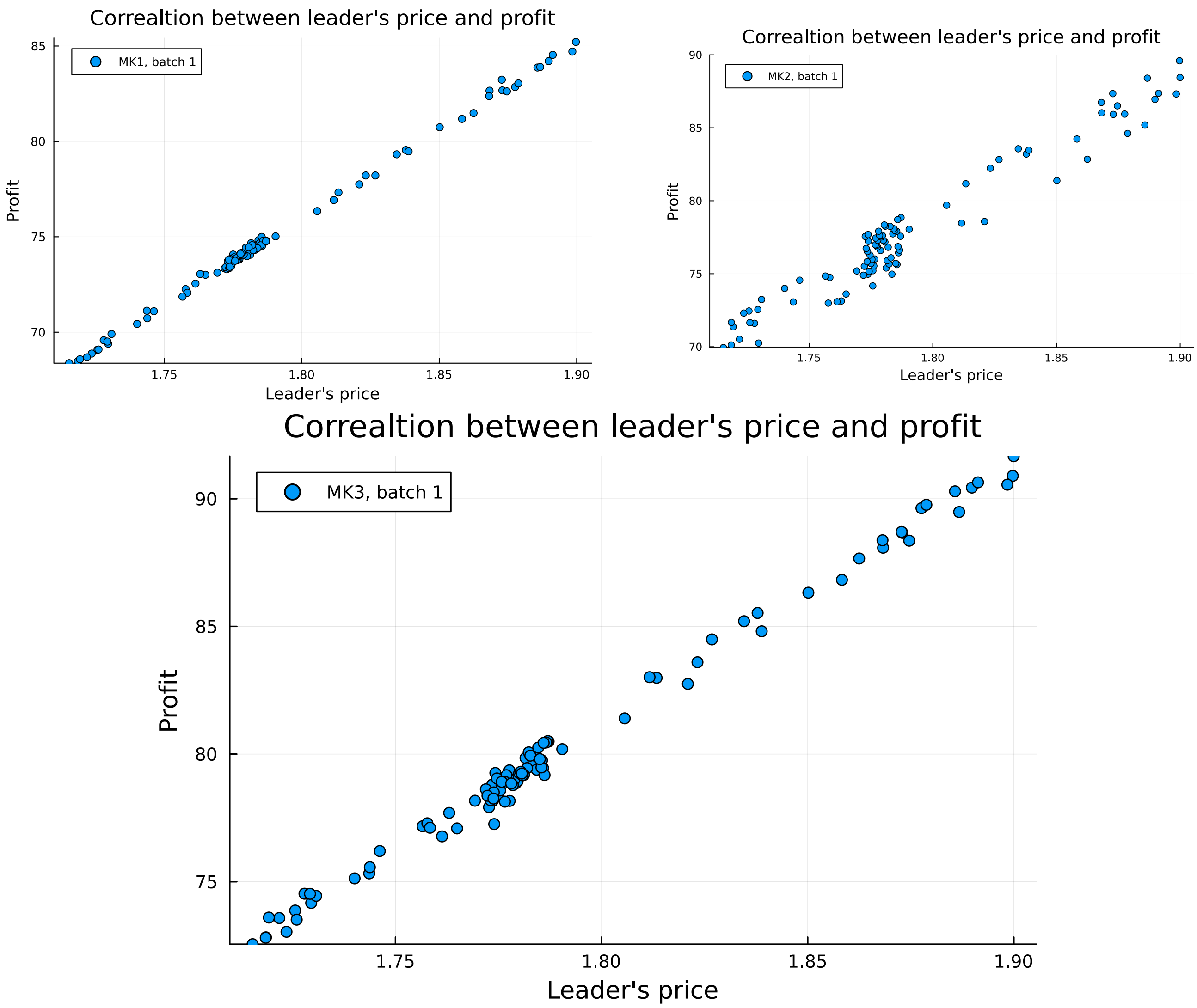


Figure 8: MK1, MK2, MK3: there is a strong correlation between profit and leader's price, within the intervals graphed.

Explore phase

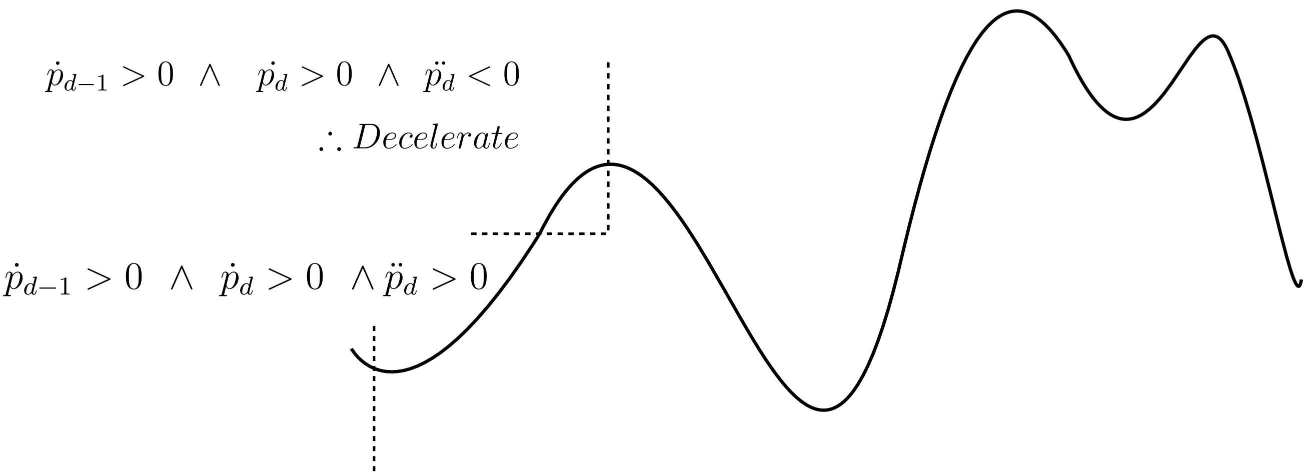
During the exploration phase, a table-based data structure is built which stores information about prices and profitability. These tables can then be used to generate numerical approximations of derivatives, based on large step sizes.
The explore phase has been implemented based on the following conditions:

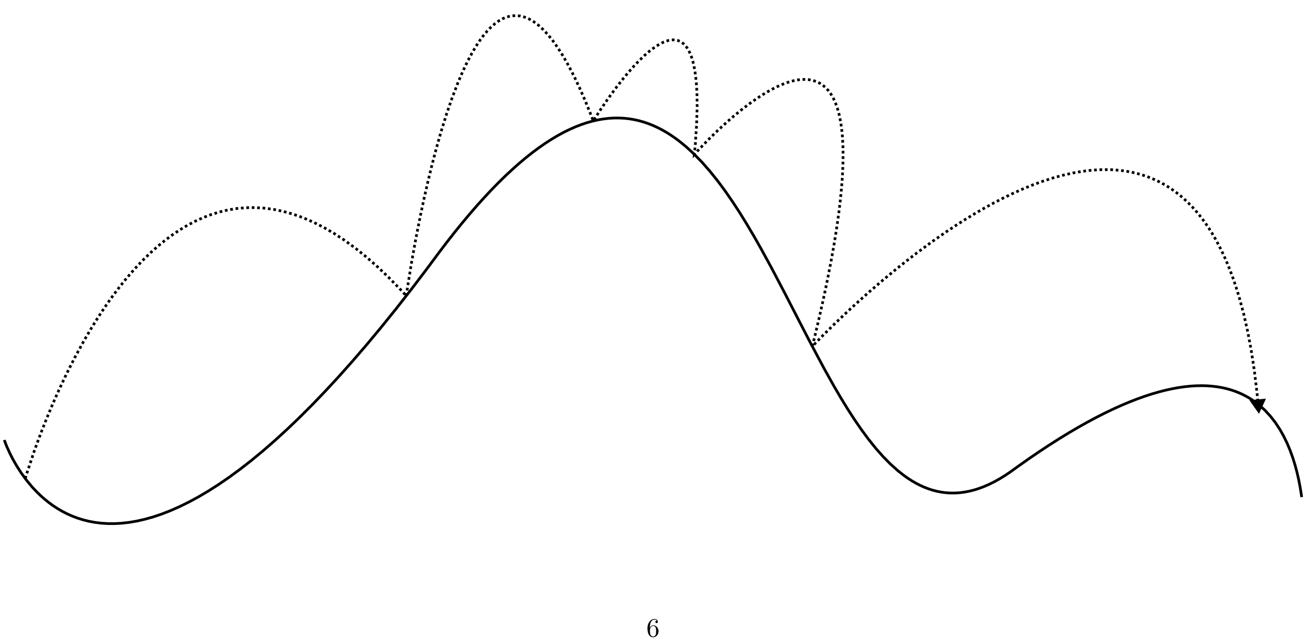
- An initial non-zero gradient is set, as a hyperparameter
- The ratio of the days dedicated to explore:exploit is a hyperparameter, typically 15:15
- If the profit becomes negative, end the exploration phase immediately
- In the context of MK3, in the case that the Leader's Price is iterated beyond the upper bound of 15, reduce it to ~14 and add noise within the bounds of $[1, 0]$ via a random number generator, to give the binary search in the exploit phase some leeway for finding an optimum
- The relationship $\frac{dP}{dD}$ is not entirely understood, but is estimated relatively negligible compared to $\frac{dP}{dL}$

The acceleration of the step sizes (velocity) of the 4 curve contexts:

1. $\frac{dP}{dL} > 0$ at d, $\frac{dP}{dL} > 0$ at $d-1$, and $\frac{d^2P}{dL^2} < 0$ for both d and $d-1$
2. $\frac{dP}{dL} > 0$ at d, $\frac{dP}{dL} < 0$ at $d-1$, and $\frac{d^2P}{dL^2} > 0$ for both d and $d-1$, where d represents the most recent date
3. $\frac{dP}{dL} < 0$ at d, $\frac{dP}{dL} < 0$ at $d-1$, and $\frac{d^2P}{dL^2} < 0$ for both d and $d-1$, where d represents the most recent date
4. $\frac{dP}{dL} < 0$ at d, $\frac{dP}{dL} > 0$ at $d-1$, and $\frac{d^2P}{dL^2} > 0$ for both d and $d-1$, where d represents the most recent date

These contexts are akin to slicing the PDF of a normal distribution into 4 quadrants. The rates of change based on the above 4 contexts were then implemented as hyperparameters that could be adjusted.



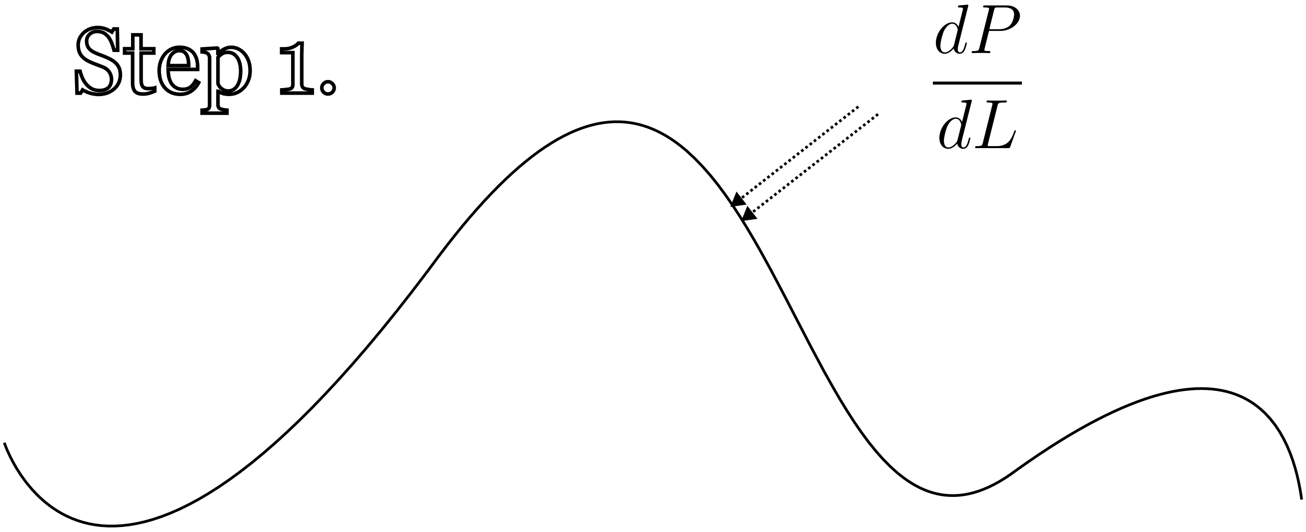


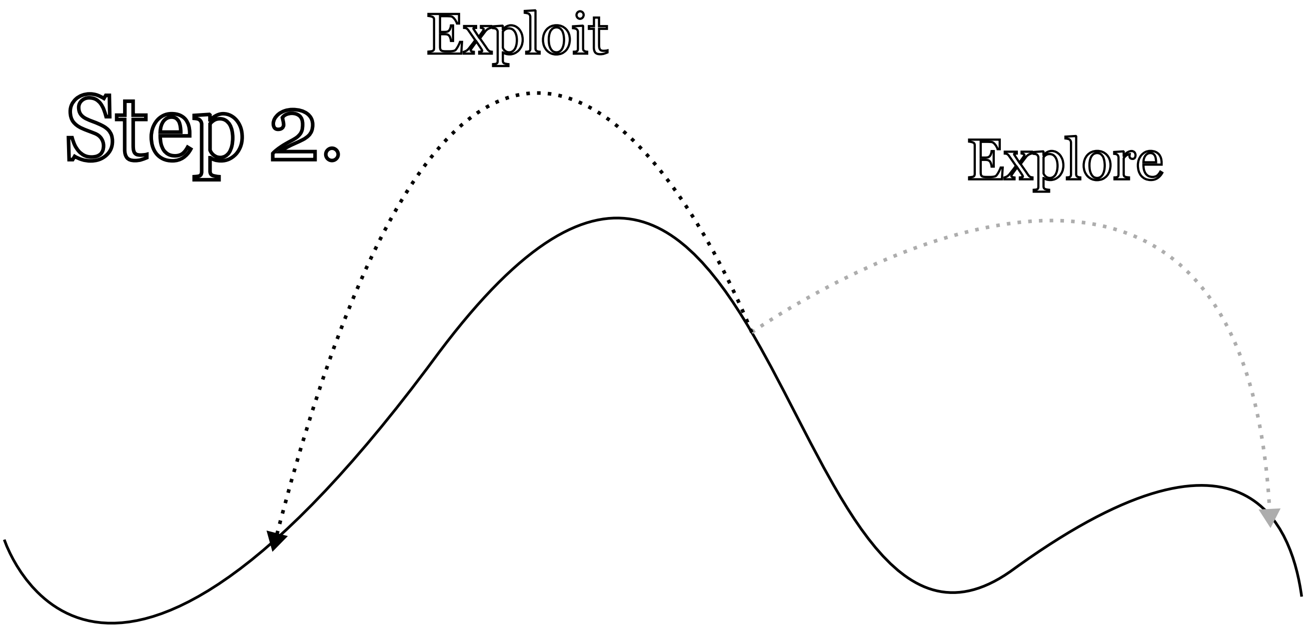
Exploit phase

The exploitation phase makes the following sequence:

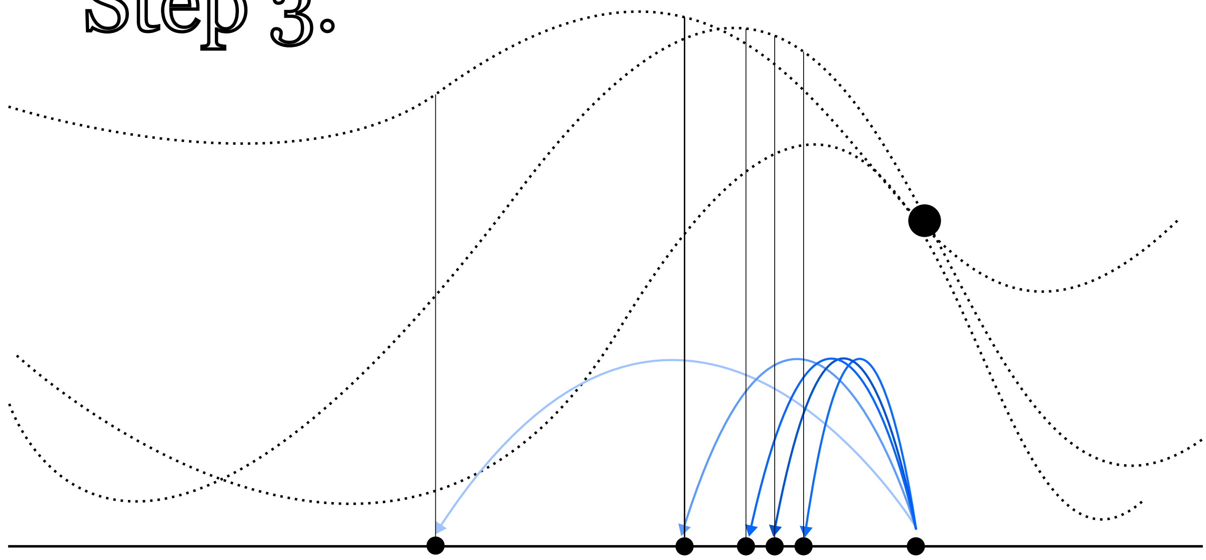
1. Take the Leader's Price which generates the highest profitability and, as the side of the curve on which it is on is not known, make a tiny increment on it, to calculate a derivative
2. Take the gradient generated during explore phase at the current known optimum point, and, if on the right-hand side of the optimum curve, negate it, else if on the left-hand side of curve, halve it
3. Because the underlying shape of the curve that is to be traversed is not known, run a binary search that factors in profitability of each Leader's Price traversed, so as to deduce maximum of curve

Because of the complexity of the binary search implementation, particularly in regards to state, interactions with the simulator, and edge cases, a fallback (via try-except pattern) has been implemented, which simply selects the best known Leader's Price during the explore phase.





Step 3.



Fine-tuning

Once the implementation was done, it was a matter of adjusting the hyperparameters to find what approach worked best. The following method tended to produce good results for MK1, MK2:

1. Context 1: Whilst climbing the curve, accelerate rapidly
2. Context 2: Whilst approaching the peak, accelerate less rapidly, or decelerate
3. Context 3: After passing the peak, decelerate
4. Context 4: When approaching a local minimum, accelerate

The above was then combined with a short explore phase of only 6 days, and a long exploit phase.

As for MK3, the approach was similar to the above, but with slower acceleration, to ensure not to hit the upper limit.

A small initial speed (”initial boost”) was preferred, to reduce dependency on this hyperparameter, so that the implementation would be more adaptable to unknown curves, via the 2nd derivative.

The exact numerical table of hyperparameters can be found in the associated Jupyter notebook.

Future potential

If the ideas could somehow be extrapolated to a multidimensional context, they would be applicable to an optimiser of a probabilistic neural network (non-trivial task).

References

[BEKS17] Jeff Bezanson, Alan Edelman, Stefan Karpinski, and Viral B Shah. Julia: A fresh approach to numerical computing. *SIAM Review*, 59(1):65–98, 2017.